B.Tech CSE - Third Year | Lab 4

Assignment 4: Selection Sort • *Aim: Sort arrays by selecting the minimum from the unsorted part.*

Experiment No. 4.1: Find index of smallest value

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
public class Task_01_Find_index_of_smallest_value {
    public static void main(String[] args) {
        int[] a = {64, 25, 12, 22, 11};
        int minIndex = 0;
        for (int i = 1; i < a.length; i++) {
            if (a[i] < a[minIndex]) minIndex = i;
        }
        System.out.println(minIndex);
    }
}
```

3. **Output:**

```
4
```

4. **Viva-Voce:** What does an index represent? Why store minIndex rather than only minimum value? What is the index of 11?

Experiment No. 4.2: Sort an array using selection sort

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
public class Task_02_Sort_an_array_using_selection_sort {
    public static void main(String[] args) {
        int[] a = {64, 25, 12, 22, 11};
        for (int i = 0; i < a.length - 1; i++) {
            int minIndex = i;
            for (int j = i + 1; j < a.length; j++) {
                if (a[j] < a[minIndex]) minIndex = j;
            }
            int temp = a[i]; a[i] = a[minIndex]; a[minIndex] = temp;
        }
        for (int value : a) System.out.println(value);
    }
}
```

3. **Output:**

```
11
12
22
25
64
```

4. **Viva-Voce:** What is sorted and unsorted part? How does selection sort choose each next value? Is selection sort in-place?

Experiment No. 4.3: Perform first selection-sort step

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
public class Task_03_Perform_first_selection {
    public static void main(String[] args) {
        int[] a = {64, 25, 12, 22, 11};
        int minIndex = 0;
        for (int i = 1; i < a.length; i++) {
            if (a[i] < a[minIndex]) minIndex = i;
        }
        int temp = a[0]; a[0] = a[minIndex]; a[minIndex] = temp;
        for (int value : a) System.out.println(value);
    }
}
```

3. **Output:**

```
11
25
12
22
64
```

4. **Viva-Voce:** What is the first step of selection sort? Which values are swapped here? Is the array fully sorted after one step?